

Industrial Electrical Safety & Compliance Overview

A working reference for plant owners, project managers and EPC contractors who need every electrical activity to be statutorily compliant, audit-ready and safe.

Version 1.0 · India — CEA, Factories Act, IS/IEC framework

1. Governing Statutes & Standards

- Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023
- Indian Electricity Rules and State Electricity Inspectorate notifications
- Factories Act 1948 — Chapter IV (Safety) and applicable State Factory Rules
- IS 3043 (Earthing), IS 5216 (Safety procedures), IS/IEC 60364 (LV installations)
- IS/IEC 61936 (HV installations > 1 kV), IS/IEC 60909 (short-circuit calculations)
- OISD-173 / OISD-189 for hydrocarbon and process industries

2. Statutory Approvals VECTREV Handles

- CEIG / Electrical Inspectorate drawing approval and pre-energisation inspection
- DG set installation approval and PESO clearance (where applicable)
- Periodic statutory testing of earth pits, lightning arresters and HV equipment
- Energy audit and mandatory BEE compliance reports for designated consumers

3. On-Site Safety Protocols

- JSA and HIRA before every shift
- Permit-to-Work system: hot work, height, confined space, electrical isolation
- Lock-Out / Tag-Out (LOTO) on every isolation point with personal padlocks
- Arc-flash boundary calculation and PPE matrix (Cat 1-4) per IEEE 1584
- Live-line work prohibited except under written sanction
- Toolbox talks recorded daily; incident reporting within 24 hours

4. Documentation Plant Auditors Expect

- Electrical safety policy, organisation chart and competent person register
- Calibrated test instrument register with traceable certificates
- Earth resistance, IR and HV test records with periodicity and trending
- Statutory Form-E / Form-O reports issued by licensed electrical inspector
- Training matrix, first-aider list and refresher records
- Investigation reports for near-misses, electrical incidents and failures

5. PPE & Safety Equipment Baseline

- Class 0 / Class 2 insulated gloves with annual dielectric test
- Arc-rated coveralls, balaclava and face shield matched to incident energy
- Insulated tools, voltage detector and rubber mat at every panel front
- Rescue hook, shorting and earthing equipment on HV switchyards
- Approved fire extinguishers (CO2 / Clean Agent) at electrical rooms

6. Compliance Audit Scope (VECTREV)

- Walk-down of all electrical rooms, substations and DG yards
- Sample testing of earthing, IR and protection settings against design
- Document review against statutory and IS standards
- Gap report with prioritised actions, owners and target dates
- Re-audit and closure verification with photo evidence

7. Engagement Models

Project-based: single-site safety audit and compliance dossier preparation.

Retainer: quarterly site visits, statutory liaison and incident response.

Embedded engineer: dedicated on-site safety lead for shutdowns and major projects.

Speak to a compliance engineer

Call **+91 88796 08428** or email **contact@vectrev.in**. VECTREV will walk your site, identify gaps and return a prioritised compliance roadmap within two weeks.